

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0702

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

*I K. Yamini (192572181) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

K. Yamini

20/20
20/20

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.
3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding ; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method

Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1. subnetting of 192.168.20.0/24 for three Labs

Problem understanding

The given network 192.168.20.0/24 must be divided into three subnets using fixed subnet masks:

Lab A $\rightarrow$ /26, Lab B $\rightarrow$ /27, Lab C $\rightarrow$ /28.

we must find network address, host range, broadcast address, and usable hosts.

method selection

use subnetting formulas:

- Block size = $2^{(32 - \text{subnet})}$
- usable hosts = $2^{(32 - \text{subnet})} - 2$

solution process

Lab A (/26)

• subnet mask: 255.255.255.192

• Block size = 64

Network Address: 192.168.20.0

Broadcast Address: 192.168.20.63

Host Range: 192.168.20.1 - 192.168.20.62

Usable Hosts: 62

Lab B (/27)

• subnet mask: 255.255.255.224

• Block size = 32

Network Address: 192.168.20.64

Broadcast Address: 192.168.20.95

Host Range: 192.168.20.65 - 192.168.20.94

Usable Hosts : 30

Lab C (/28)

• subnet mask : 255.255.255.240

• Block size = 16

Network Address = 192.168.20.96

Broadcast Address = 192.168.20.111

Host Range : 192.168.20.97 - 192.168.20.110

usable Hosts : 14

Final Answer

Lab A $\rightarrow$ 192.168.20.0/26 (62 hosts)

Lab B $\rightarrow$ 192.168.20.64/27 (30 hosts)

Lab C $\rightarrow$ 192.168.20.96/28 (14 hosts)

2. subnet Analysis of Two systems

Problem understanding

Two systems are given with IP addresses and subnet mask.
we must find the subnet details and verify if IP's are valid.

method selection

For /26 :

• Block size = 64

• subnets : 0-63, 64-127, 128-191, 192-255

solution process

system 1 : 192.168.1.130

Falls in subnet 128-191

Network Address : 192.168.1.128

Broadcast Address : 192.168.1.191

Host Range : 192.168.1.129 - 192.168.1.190

Valid Host : yes

System 2 : 192.168.1.190

Falls in same subnet 128-191

Network Address : 192.168.1.128

Broadcast Address : 192.168.1.191

Host Range : 192.168.1.129 - 192.168.1.190

Valid Host : yes

Final Answer

Both systems belong to the same subnet 192.168.1.128/26 with 62 usable hosts.

3. Subnetting of 172.16.0.0/16 for Departments

Problem understanding

The network must be divided among HR, sales, IT, and Admin using different subnet sizes.

Method selection

Allocate subnets sequentially using given prefix lengths.

Solution process

sales
HR (126)

Network Address : 172.16.0.128

Broadcast Address : 172.16.0.191

Host Range : 172.16.0.129 - 172.16.0.190

Usable Hosts : 62

HR (125)

Network Address : 172.16.0.0

Broadcast Address : 172.16.0.127

Host Range : 172.16.0.1 - 172.16.0.126

Usable Hosts : 126

IT (127)

Network Address : 172.16.0.192

Broadcast Address : 172.16.0.223

Host Range : 172.16.0.193 - 172.16.0.222

Usable Hosts : 30

Admin (128)

Network Address : 172.16.0.224

Broadcast Address : 172.16.0.239

Host Range : 172.16.0.225 - 172.16.0.238

Usable Hosts : 14

Remaining Address Range

172.16.0.240 - 172.16.255.255

Final Answer

HR : 126 hosts

Sales : 62 hosts

IT : 30 hosts

Admin : 14 hosts

subnetting 172.16.10.0/24 using /27

problem understanding

we must calculate subnets, hosts per subnet and identify subnet.

method selection

- subnets = $2^3 = 8$
- Hosts = $2^5 - 2 = 30$
- Block size = 32

solution process

subnets :

0-31, 32-63, 64-95, 96-127, ... etc.

IP 172.16.10.78

Falls in 64-95

Network Address : 172.16.10.64

Broadcast Address : 172.16.10.95

Host Range : 172.16.10.65 - 172.16.10.94

IP 172.16.10.95

It is broadcast address, not a valid host.

But it belongs to same subnet.

Final Answer

Total subnets : 8

Hosts Per subnet : 30

subnet of IP 78 : 172.16.10.64/27

5. IPv6 Address Analysis

Problem understanding

we must compress, expand, and find network prefix & no. of hosts.

Method selection

Apply IPv6 rules:

- Remove leading zeros
- Replace consecutive zeros with ::

Solution process

compressed Address

2001:db8::ff00:42:8329

Expanded Address

2001:0db8:6000:0000:0000:ff00:0042:8329

Network Prefix (/64)

2001:db8:0:0::/64

Number of Hosts

$$2^{64} = 18,446,744,073,709,551,616$$

Final Answer

compressed : 2001:db8::ff00:42:8329

Expanded : 2001:0db8:0000:0000:0000:ff00:0042:8329

Network Prefix : 2001:db8:0:0::/64

Hosts : 2^{64}